

1. I started out making a simple compute shader renderer and then started thinking about what I would make. I kinda wanted to implement this paper I saw that let you efficiently render an implicit surface without any lossy meshing (marching cubes) or that trick in raymarching that costs you a lot of performance if you're not rendering an SDF. The issue is that that approach to rendering on its own struggles to be combined with anything that lets you do global illumination. The further work section described a direction I might take this in the future though.

So I went with classic raymarching. I tried to put together something somewhat Cornell Box looking, but because the light at the top would be quite annoying to put together the SDF of so instead I made the two walls emissive and made the rest of the walls only slightly emissive but mostly they pick up the lighting. It took a bit to make the random number generation seed actually look random and properly time varying but eventually I got it. I used that random seed to add lambertian bouncing. There is only one bounce, but it still looks really good in my opinion. I then added the ability to accumulate lighting into an image to average the randomness between frames. Also, I made it possible to press shift to zero out the accumulated buffer and just send the current frame directly to the screen without accumulation. I decided to generate the ray directions using a PV matrix so it was a lot easier to move the camera around and rotate it. So I made the camera able to move as well. I then added a fractal-ish shape by using an octahedron SDF and mirroring and shifting and rotating the point where the SDF is sampled in a loop. I then connected some of the rotation parameters to a frame counter. I kinda like the way that the fractal moves in between frame accumulations. It looks pretty neat in my opinion.

I was aiming for a proper 3d look with realistic lighting. I ended up with a lot more noise in the image than I would've liked. I think its because the way I generate the RNG seed is still a bit weird. But thats not an overall too big problem in my opinion. I think the lighting looks awesome though. I'm pretty proud of it.